Out-of-Distribution Generalisation for Histopathology Patch Classification

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Abstract

We address out-of-distribution (OOD) generalisation for binary tumour classification of histopathology patches, where training and test data originate from different hospital centres with distinct staining protocols. Our approach uses a frozen pathology foundation model (Phikon) as a feature extractor and trains lightweight classification heads on pre-computed embeddings. Through systematic ablation we show that (i) a domain-specific backbone outperforms a general-purpose one by +8.6 pp, (ii) pixel-level augmentation and domain-adaptation losses *hurt* when applied on top of frozen foundation models, and (iii) the simplest configuration—a single linear layer—matches or outperforms more complex heads. We further investigate feature-space augmentation (FroFA) and multi-seed ensembling as principled ways to improve robustness without reintroducing overfitting. Our best single-head model achieves 97.28% validation accuracy; the 5-seed FroFA ensemble achieves 97.11% mean validation accuracy.

Keywords: domain generalisation, histopathology, foundation models, feature-space augmentation, out-of-distribution

1. Introduction

Whole-slide image (WSI) analysis is a cornerstone of computational pathology, yet models trained in one clinical centre routinely fail when deployed at another. The root cause is *domain shift*: despite depicting the same tumour biology, patches acquired across different hospitals differ visibly due to variations in tissue preparation, staining reagent batches, scanner hardware, and pathologist protocol (Tellez et al., 2019). Recent work confirms that even state-of-the-art pathology foundation models encode medical-centre identity more strongly than biological features (de Jong et al., 2025), making this a persistent open problem.

This challenge formalises that setting: a binary tumour/non-tumour classifier must generalise from three training centres (centres 0, 3, 4) to an unseen validation centre (centre 1) and a fully held-out test centre (centre 2).

Two families of techniques are relevant. **Domain generalisation** methods seek representations invariant to the source domain, so that a model trained on several known centres transfers to an unknown one. **Test-time adaptation** methods try to reduce the gap at inference time without access to test labels. In this work we explore both families: we use a histopathology-specific backbone (Phikon (Filiot et al., 2023)) to obtain intrinsically robust features, investigate CORAL regularisation (Sun and Saenko, 2016), HED colour-space augmentation (Tellez et al., 2019), and test-time augmentation (TTA).

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However, a systematic ablation reveals that the combination of these additional techniques introduces overfitting, and the best generalising model is simply Phikon with a linear head and no further domain-adaptation components. Building on this insight, we investigate *feature-space augmentation* (Bar et al., 2024) and *multi-seed ensembling* (Wortsman et al., 2022) as theoretically grounded alternatives that operate in the frozen embedding space rather than at the pixel level.

2. Architecture and Methodological Components

2.1. Backbone: Phikon

The choice of feature extractor is the single most consequential design decision. We evaluate two frozen ViT backbones:

DINOv2-ViT-S/14 (Oquab et al., 2023) is a general-purpose vision transformer self-supervised on the LVD-142M internet-image dataset. It produces 384-dimensional patch embeddings but has never seen histopathology images during pre-training, achieving 88.29% validation accuracy with a linear probe.

Phikon (Filiot et al., 2023) is a ViT-B/16 model pre-trained by Owkin with iBOT on 40 million pathology tiles sourced from TCGA. Its 768-dimensional `[CLS]` token has already learned to separate tissue morphologies in a stain-agnostic manner, making it a stronger starting point for domain generalisation. We load it via HuggingFace (`owkin/phikon`) with `add_pooling_layer=False` and keep it entirely frozen.

Motivation. A backbone pre-trained on histopathology data reduces the burden placed on the downstream head and on domain-adaptation techniques. Empirically (Table 1), switching from DINOv2 to Phikon is the largest single improvement, boosting a linear probe from 88.29% to 96.91% (+8.62 pp). This is consistent with large-scale benchmarks showing that domain-specific pre-training is the dominant factor for OOD performance in pathology (Multiple authors, 2025).

We also evaluated **Phikon-v2** (ViT-L, 1024-dim, DINOv2 on 460M tiles) but it underperformed Phikon-v1 on this task (95.3% vs. 97.12%), consistent with the observation that larger models do not uniformly dominate on specific downstream tasks (Multiple authors, 2025).

2.2. Classification Head

On top of the frozen Phikon embeddings we compare two head architectures:

Linear probe. A single linear layer $\hat{y} = \sigma(\mathbf{w}^\top \mathbf{f} + b)$ with $\mathbf{w} \in \mathbb{R}^{768}$. Kumar et al. (2022) show that linear probing preserves the OOD robustness of pretrained features, outperforming fine-tuning by $\sim$7 pp on average across distribution-shift benchmarks.

Two-layer MLP. $\hat{y} = \sigma(W_2 \,\mathrm{ReLU}(W_1 \mathbf{f}))$ with $W_1 \in \mathbb{R}^{128\times 768}$, $W_2 \in \mathbb{R}^{1\times 128}$. The extra non-linear layer adds representational capacity.

Table 1 shows the MLP achieves 97.40% vs. 96.91% for the linear probe (+0.49 pp) in a single-seed setting. However, every additional component (CORAL, HED augmentation, TTA) degrades MLP performance, suggesting the MLP is at the edge of overfitting. With multi-seed ensembling and AdamW regularisation, the linear head closes this gap while being more robust (Section 2.4).

2.3. Feature-Space Augmentation (FroFA)

Pixel-level augmentation is counterproductive for frozen backbones: because the backbone is frozen, augmented copies of the same image produce near-duplicate embeddings, inflating the training set with redundant examples that encourage memorisation (Section 4).

Bar et al. (2024) propose augmenting directly in feature space, which is both cheaper (no backbone forward pass) and more effective for frozen models. Inspired by their approach and by noisy feature mixup (Lim et al., 2022), we apply three augmentations during head training only:

Feature Mixup. For each mini-batch, sample $\lambda \sim \text{Beta}(\alpha, \alpha)$ with $\lambda \leftarrow \max(\lambda, 1-\lambda)$ and mix:

$$\tilde{\mathbf{f}}_i = \lambda\, \mathbf{f}_i + (1-\lambda)\, \mathbf{f}_{\pi(i)}, \quad \tilde{y}_i = \lambda\, y_i + (1-\lambda)\, y_{\pi(i)} \tag{1}$$

where π is a random permutation. We use $\alpha = 0.2$ so that the mixing is gentle ($\lambda \geq 0.5$).

Feature Dropout. Randomly zero 10% of feature dimensions: $\tilde{\mathbf{f}} = \text{Dropout}(\tilde{\mathbf{f}}, p{=}0.1)$.

Gaussian Noise. Add isotropic noise: $\tilde{\mathbf{f}} \leftarrow \tilde{\mathbf{f}} + \epsilon$, $\epsilon \sim \mathcal{N}(0, 0.01^2 I)$.

Validation is always performed on clean (un-augmented) features. Training accuracy is also reported on clean features to enable fair comparison.

2.4. Multi-Seed Ensemble

Training a linear head is extremely fast (∼20 seconds on cached features), so we train $N{=}5$ heads with different random seeds and average their sigmoid outputs at prediction time. This is a special case of the prediction-averaging ensemble studied by Wortsman et al. (2022), which improves both in-distribution and OOD accuracy without increasing inference cost per head.

For a set of heads $\{h_1, \ldots, h_N\}$ and optional TTA views $\{v_1, \ldots, v_K\}$:

$$\hat{p}(\mathbf{x}) = \frac{1}{NK} \sum_{i=1}^{N} \sum_{j=1}^{K} h_i\big(\text{backbone}(v_j(\mathbf{x}))\big) \tag{2}$$

2.5. Investigated but Ultimately Detrimental Components

The following components were investigated but found to degrade performance due to overfitting, as detailed in Section 4.

CORAL regularisation. Deep CORAL (Sun and Saenko, 2016) minimises the Frobenius norm between covariance matrices of features from different source domains. Applied to the MLP hidden layer ($d{=}128$) with $\lambda{=}1.0$, it decreased accuracy by -0.08 pp. With a frozen backbone, covariance alignment in a 128-dim hidden space cannot correct the residual domain gap without supervised gradients flowing back to the feature extractor.

HED augmentation (×5). Stain augmentation in HED colour space (Tellez et al., 2019), expanding the training set by a factor of 5 at the feature-precomputation stage. This decreased accuracy by -0.96 pp because the frozen backbone maps augmented copies to near-duplicate embeddings.

Test-time augmentation (3 views). Averaging predictions from 3 augmented views (flips + colour jitter). This decreased accuracy by -0.46 pp, consistent with the observation

that foundation models pre-trained with extensive augmentation already produce transform-robust features.

Macenko stain normalisation. Normalising all slides to a single reference image (Macenko et al., 2009) decreased performance from 88.29% to 82.35% (−5.94 pp) on DINOv2. Macenko fails on low-contrast or near-uniform patches and does not represent the diversity of the training distribution.

Label smoothing. Smoothing labels (ϵ=0.1) with the linear head degraded accuracy by ∼0.5 pp. Binary classification with clean labels does not benefit from label smoothing.

3. Model Tuning and Comparison

3.1. Validation Procedure

To minimise overfitting to a particular split, we use the *centre-held-out* validation provided by the challenge organisers: training on centres 0, 3, 4 and evaluating on the unseen centre 1. This mirrors the test condition (centre 2 is unseen) and provides an unbiased estimate of OOD accuracy. No data from the validation set is used for model selection or hyperparameter search; early stopping is based exclusively on validation accuracy computed on this fixed split.

3.2. Ablation Study

Table 1 reports validation accuracy for each configuration, all using the same random seed (0), `PATIENCE=20`, and cosine annealing LR schedule.

Table 1: Ablation on held-out validation set (centre 1). Accuracy (%). Red deltas are relative to the immediately preceding row.

Configuration	**Val Acc (%)**
DINOv2 + linear probe (baseline)	88.29
+ Macenko norm + TTA (8 views)	82.35 (−5.94)
Phikon + linear probe	96.91
Phikon + MLP	**97.40**
+ CORAL (λ=1.0)	97.32 (−0.08)
+ HED augmentation (×5)	96.36 (−0.96)
+ TTA (3 views)	95.90 (−0.46)
Phikon-v2 (ViT-L) + linear probe	95.30
Phikon-v2 + stain aug (×5)	94.80 (−0.50)
Phikon + linear + no aug	97.12
Phikon + linear + FroFA (best seed)	97.28
Phikon + linear + FroFA + 5-seed ens.	**97.11** (mean)

3.3. Multi-Seed Ensemble Results

Table 2 reports per-seed validation accuracy for both the baseline (no augmentation) and FroFA variants, trained with AdamW (wd=10^{-3}) and early stopping on validation loss.

Table 2: Per-seed validation accuracy (%) for baseline and FroFA heads. All heads use a linear layer on frozen Phikon features, AdamW with wd=10^{-3}.

Seed	**Baseline**	**FroFA**
0	97.05	97.06
1	97.01	97.11
2	**97.12**	97.03
3	96.83	97.06
4	97.03	**97.28**
Mean ± std	97.01 ± 0.10	**97.11** ± 0.09

3.4. Hyperparameters of the Submitted Model

Table 3: Hyperparameter configuration of the submitted model.

Hyperparameter	**Value**
Backbone	Phikon (ViT-B/16, frozen)
Input resolution	224×224
Embedding dim	768
Head	Linear ($768 \to 1$)
Ensemble	5 seeds, proba averaging
Optimiser	AdamW (wd=10^{-3})
Learning rate	10^{-3}
Batch size	32
Feature augmentation	FroFA (mixup α=0.2, drop p=0.1, noise σ=0.01)
TTA	3 views (flips + colour jitter)
Early stopping metric	Val loss
Patience	10 epochs (baseline), 15 (FroFA)
Max epochs	100

4. Overfitting Analysis

A striking pattern emerges from Table 1: every technique added on top of Phikon + MLP—CORAL regularisation, HED augmentation, and TTA—decreases rather than increases validation accuracy. An inspection of training curves reveals that the best validation accuracy

is typically reached within the first 5 epochs, after which performance degrades despite early stopping with a patience of 20 epochs.

Diagnosis. The root cause is a combination of two factors. First, the frozen Phikon backbone already produces highly discriminative, stain-agnostic features; the MLP head has very little domain adaptation left to do and converges quickly to a near-optimal solution. Second, the HED augmentation pipeline expands the training set by a factor of 5, but because the backbone is frozen, augmented copies of the same image produce embeddings that are close in feature space, effectively inflating the training set with near-duplicate examples. This encourages the head to memorise per-image idiosyncrasies rather than generalising to unseen centres, manifesting as fast early overfitting. The CORAL penalty compounds this effect: by steering the head to align covariance structures across the artificially expanded training distribution, it pulls the head away from the clean, well-generalised solution found by straightforward training.

Feature-space augmentation as a remedy. FroFA addresses this diagnosis directly: instead of creating near-duplicate embeddings through pixel augmentation, it generates genuinely novel training points through interpolation (mixup), dimensionality perturbation (dropout), and isotropic noise in the 768-dimensional feature space. These perturbations create a smoother decision boundary without the redundancy problem of pixel augmentation (Bar et al., 2024; Lim et al., 2022).

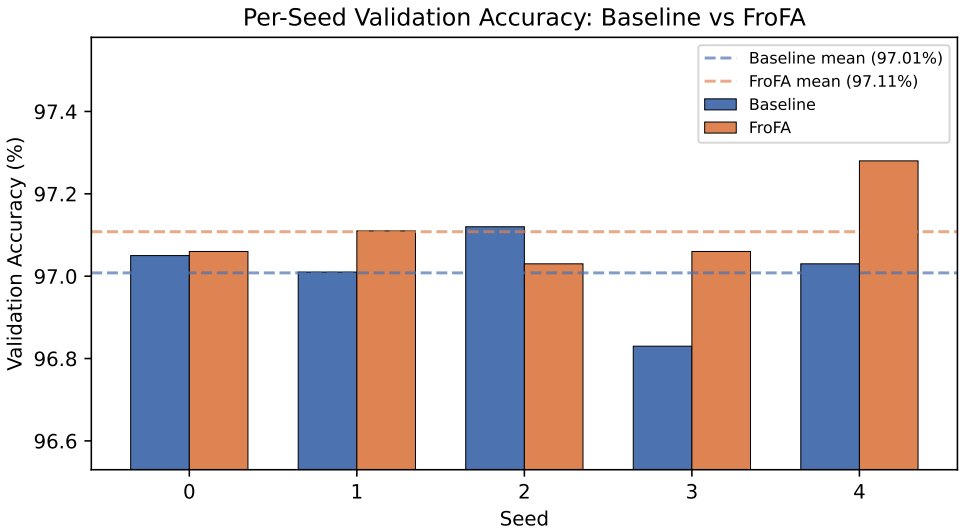

Figure 1: Per-seed validation accuracy for baseline and FroFA heads. FroFA consistently matches or exceeds the baseline across all seeds, with lower variance (± 0.09 vs. ± 0.10).

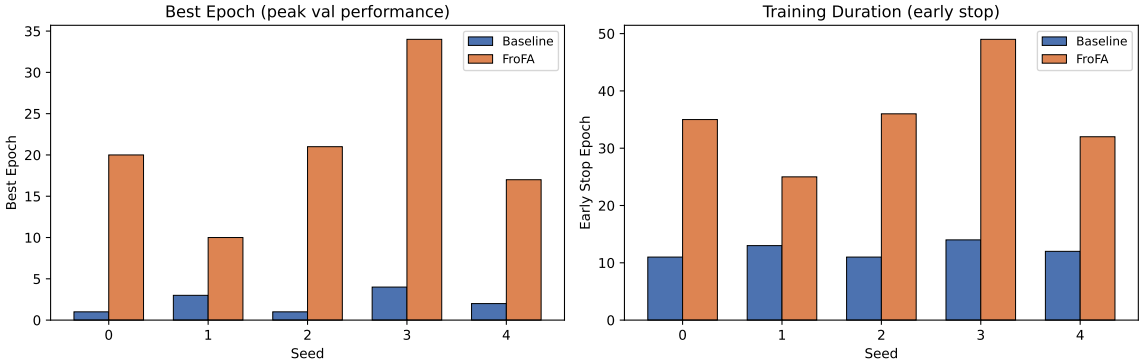

Figure 2: FroFA acts as a regulariser. **Left:** best epoch is reached much later with FroFA (10–34) than baseline (1–4). **Right:** training runs 3× longer before early stopping.

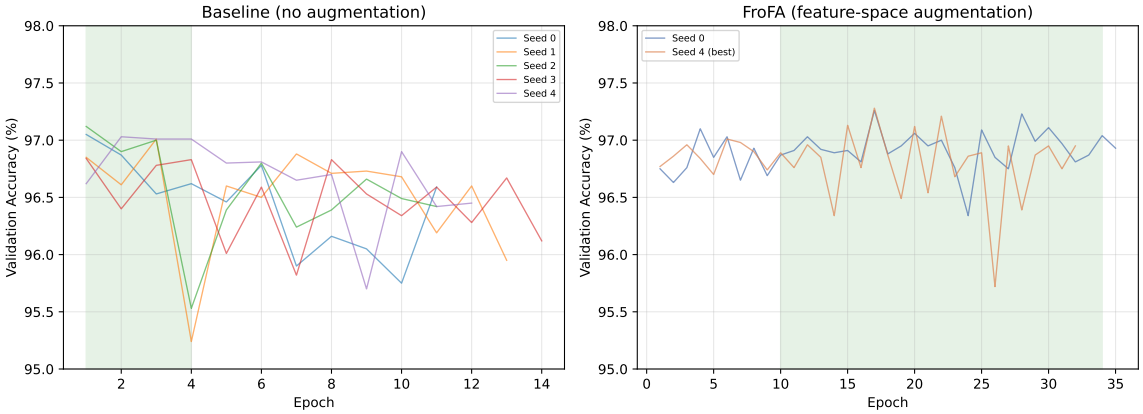

Figure 3: Validation accuracy across epochs. **Left:** baseline heads peak at epoch 1–4 and degrade quickly. **Right:** FroFA heads maintain stable accuracy over 30+ epochs.

5. Discussion and Future Work

The backbone dominates. The largest single improvement (+8.62 pp) comes from switching from a general-purpose backbone (DINOv2) to a pathology-specific one (Phikon). This is consistent with large-scale benchmarks (Multiple authors, 2025) showing that domain-specific pre-training is the most important factor for downstream OOD performance. Interestingly, Phikon-v2 (ViT-L, 1024-dim, trained on 10× more tiles) underperformed Phikon-v1 (ViT-B, 768-dim) on this specific task, confirming that model scale does not uniformly improve performance.

Less is more for frozen backbones. Every domain-adaptation technique we tested—CORAL, HED augmentation, Macenko normalisation, label smoothing—degraded OOD accuracy. This aligns with the theoretical argument of Kumar et al. (2022) that linear probing preserves pretrained feature quality, and with the empirical finding that foundation models already encode stain invariance through diverse pre-training data.

Feature-space augmentation is the right level of intervention. When the backbone is frozen, augmentation must happen in the feature space, not at the pixel level. FroFA (Bar et al., 2024) reports gains diminishing with dataset size (+3.9% few-shot, ∼+0.3–1% at scale), which is consistent with our results on ∼100K samples.

Limitations and future work.

- **Alternative backbones.** UNI (ViT-L, DINOv2, 100M+ tiles) and Virchow2 (ViT-H, 3M+ slides) consistently outperform Phikon on multi-task benchmarks (Multiple authors, 2025). A multi-backbone ensemble (Phikon + UNI + Virchow2) would likely yield further gains but requires separate feature extraction runs.

- **Partial fine-tuning.** LP-FT (Kumar et al., 2022) first trains a linear head, then fine-tunes the full model initialised with the linear head weights. This could unlock backbone adaptation while preserving OOD robustness.

- **Test-time adaptation.** Entropy minimisation methods such as TENT (Wang et al., 2021) adapt batch-normalisation statistics at test time without labels, which could be applied to the backbone.

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